Magíster en Estadística, PUCV

Statistical learning theory

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Fundamental problem in ML

Given a set of examples $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^m$ where $x_i \in \mathcal{X}$ and $y_i \in \mathcal{Y}$, find a function $h : \mathcal{X} \to \mathcal{Y}$ that:

1. Approximates well the observed data.
2. **Generalizes** to unseen data.

Fundamental assumptions

There exists an **unknown** probability distribution $P(X, Y)$ over $\mathcal{X} \times \mathcal{Y}$

Training data are **i.i.d.** samples from P:

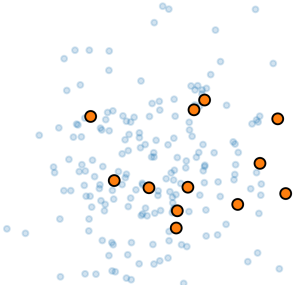
$$\{(x_1, y_1), \dots, (x_m, y_m)\} \overset{\text{i.i.d.}}{\sim} P(X, Y)$$

Hypothesis: Candidate functions $h \in \mathcal{H}$ (hypothesis space)

Fundamental assumptions

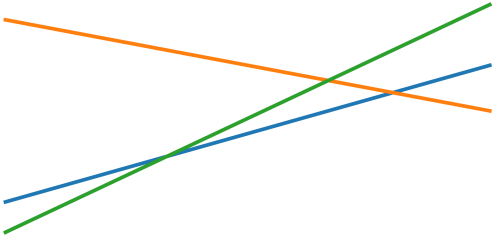
P(X,Y), training samples $\mathcal{D}$, and hypothesis space $\mathcal{H}$

Unknown distribution P(X,Y)



Training data $\mathcal{D}$
i.i.d. samples from P

Hypothesis space $\mathcal{H}$



Different hypotheses h $\in \mathcal{H}$

Loss function

Measures the **error** of an individual prediction:

$$\ell : \mathcal{Y} \times \mathcal{Y} \to \mathbb{R}_+$$

Regression:

- Squared loss: $\ell(y, \hat{y}) = (y - \hat{y})^2$
- Absolute loss: $\ell(y, \hat{y}) = |y - \hat{y}|$

Binary classification:

- 0-1 loss: $\ell(y, \hat{y}) = \mathbb{1}_{y \neq \hat{y}}$
- Log-loss: $\ell(y, p) = -y \log(p) - (1 - y) \log(1 - p)$

True risk

The **true risk** or **generalization error** of a hypothesis h is:

$$R(h) = \mathbb{E}_{(X,Y)\sim P}[\ell(Y, h(X))] = \int_{\mathcal{X}\times\mathcal{Y}} \ell(y, h(x))\, dP(x, y)$$

Objective:

$$h^* = \arg\min_{h\in\mathcal{H}} R(h)$$

P is **unknown** $\implies$ $R(h)$ is not directly computable.

Empirical risk

Given a training set $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^m$, the **empirical risk** is:

$$R_{\text{emp}}(h) = \frac{1}{m} \sum_{i=1}^{m} \ell(y_i, h(x_i))$$

- It is an **estimator** of the true risk: $R_{\text{emp}}(h) \approx R(h)$.
- It is **computable** (only depends on $\mathcal{D}$).
- By the Law of Large Numbers:

$$R_{\text{emp}}(h) \xrightarrow{m \to \infty} R(h) \quad \text{(almost surely).}$$

Empirical Risk Minimization (ERM) principle

$$\hat{h} = \text{arg}\min_{h \in \mathcal{H}} R_{\text{emp}}(h) = \text{arg}\min_{h \in \mathcal{H}} \frac{1}{m} \sum_{i=1}^{m} \ell(y_i, h(x_i))$$

Choose the hypothesis that best fits the training data.

Examples:

- Ordinary least squares (linear regression).
- Maximum likelihood.
- Classification loss minimization.

The generalization gap

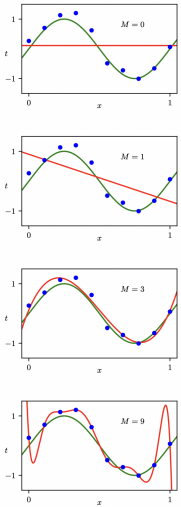
For a hypothesis $\widehat{h}$ obtained by ERM:

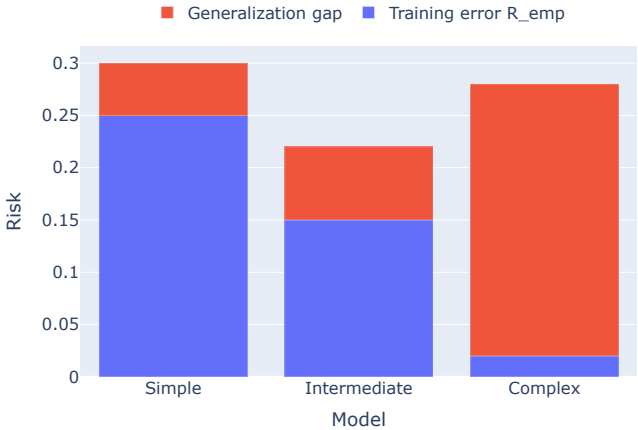
$$\text{Generalization Gap} = R(\widehat{h}) - R_{\text{emp}}(\widehat{h})$$

Error decomposition:

$$R(\widehat{h}) = \underbrace{R_{\text{emp}}(\widehat{h})}_{\text{Training error}} + \underbrace{[R(\widehat{h}) - R_{\text{emp}}(\widehat{h})]}_{\text{Generalization gap}}$$

The generalization gap





Overfitting

A model **overfits** when:

- It captures spurious patterns from the training set (noise).
- It has excellent training performance but poor generalization.

$$R_{\text{emp}}(\hat{h}) \ll R_{\text{emp}}(h^*) \quad \text{but} \quad R(\hat{h}) \gg R(h^*),$$

where h^* is a simpler hypothesis with better balance.

Capacity and Complexity of $\mathcal{H}$

Larger hypothesis spaces can overfit more easily.

$$\text{High complexity} \implies \text{Greater capacity for overfitting}$$

Complexity measures:

- **VC dimension** (Vapnik-Chervonenkis): maximum number of points that $\mathcal{H}$ can "shatter''.
- **Rademacher complexity**: average capacity to fit random labels.
- **Number of parameters**: in parametric models.

Generalization bounds (VC)

Theorem (Vapnik-Chervonenkis): With probability at least $1 - \delta$:

$$R(h) \leq R_{\text{emp}}(h) + \sqrt{\frac{8 d_{VC} \left(\ln \frac{2m}{d_{VC}} + 1 \right) + 8 \ln \frac{4}{\delta}}{m}},$$

where d_{VC} is the VC dimension of $\mathcal{H}$ and m is the training set size.

Generalization bounds (VC)

Implications:

- Generalization gap $\propto \sqrt{\frac{d_{VC}}{m}}$.
- We need $m = O(d_{VC})$ examples to learn.
- Trade-off: high $d_{VC} \implies$ better fit but worse bound.

Bias-variance decomposition

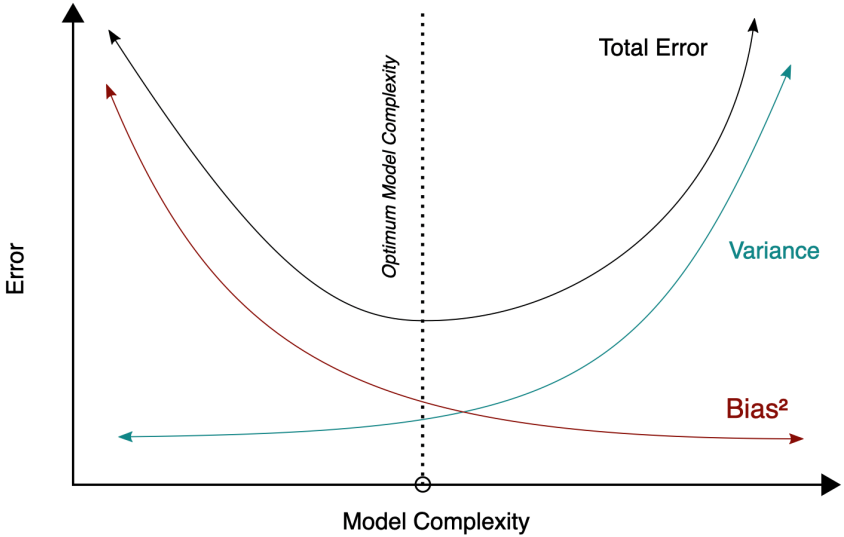
For **squared loss regression**, the expected error decomposes:

$$\mathbb{E}_{\mathcal{D}}[R(\widehat{h})] = \underbrace{\text{Bias}^2(\widehat{h})}_{\text{Approximation error}} + \underbrace{\text{Var}(\widehat{h})}_{\text{Estimation error}} + \underbrace{\sigma^2}_{\text{Irreducible noise}},$$

where:

- **Bias**: $\mathbb{E}_{\mathcal{D}}[\widehat{h}(x)] - f^*(x)$ (true target function f^*)
- **Variance**: $\mathbb{E}_{\mathcal{D}}[(\widehat{h}(x) - \mathbb{E}_{\mathcal{D}}[\widehat{h}(x)])^2]$

Bias-variance decomposition



Regularization

Modify ERM to penalize complex hypotheses:

$$\widehat{h}_\lambda = \text{arg}\min_{h \in \mathcal{H}} \left[R_{\text{emp}}(h) + \lambda \cdot \Omega(h) \right],$$

where $\Omega(h)$ is a **regularization function** and $\lambda > 0$ controls the trade-off.

Examples:

- **Ridge (L2)**: $\Omega(w) = \|w\|_2^2 = \sum_{j=1}^d w_j^2$
- **Lasso (L1)**: $\Omega(w) = \|w\|_1 = \sum_{j=1}^d |w_j|$
- **Elastic Net**: $\Omega(w) = \alpha\|w\|_1 + (1-\alpha)\|w\|_2^2$
- **Dropout, Weight Decay, Early Stopping.**

Structural Risk Minimization (SRM)

Variant of ERM that explicitly balances fit and complexity

Define hierarchy of spaces:

$$\mathcal{H}_1 \subset \mathcal{H}_2 \subset \cdots \subset \mathcal{H}_k,$$

where $d_{VC}(\mathcal{H}_1) < d_{VC}(\mathcal{H}_2) < \cdots$

Select:

$$\hat{k} = \text{arg min}_k \left[R_{\text{emp}}(\hat{h}_k) + \text{complexity}(\mathcal{H}_k, m, \delta) \right],$$

where $\hat{h}_k = \text{arg min}_{h \in \mathcal{H}_k} R_{\text{emp}}(h)$.

Model selection

Choose among multiple hypotheses/hyperparameters without overfitting in evaluation.

1. **Hold-out validation:** Split data into training/validation/test.
    - Train on $\mathcal{D}_{\text{train}}$.
    - Select model using $\mathcal{D}_{\text{val}}$.
    - Report final performance on $\mathcal{D}_{\text{test}}$.

Model selection

2. **k-fold cross-validation:**

$$\mathsf{CV}(h) = \frac{1}{k} \sum_{i=1}^{k} R_{\mathsf{emp}}^{(i)}(h),$$

where $R_{\mathsf{emp}}^{(i)}$ is error on fold i.

Information Criteria

Penalize complexity directly in a metric.

1. **Akaike Information Criterion (AIC):**

$$\text{AIC} = 2k - 2\ln(\mathcal{L}).$$

2. **Bayesian Information Criterion (BIC):**

$$\text{BIC} = k\ln(m) - 2\ln(\mathcal{L}),$$

where k is number of parameters, m sample size, $\mathcal{L}$ is likelihood.

Information Criteria

3. **Minimum Description Length (MDL):**

$$\text{MDL} = L(\text{model}) + L(\text{data}|\text{model}).$$

The curse of dimensionality

Many problems become exponentially harder with dimension.

Consequences for learning:

1. **Sparse data:** In d dimensions with m points:

   Typical distance between neighbors $\sim m^{-1/d}$.

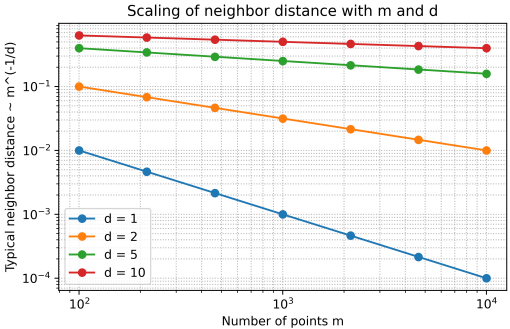
2. **Required samples:** To estimate a density with error ϵ:

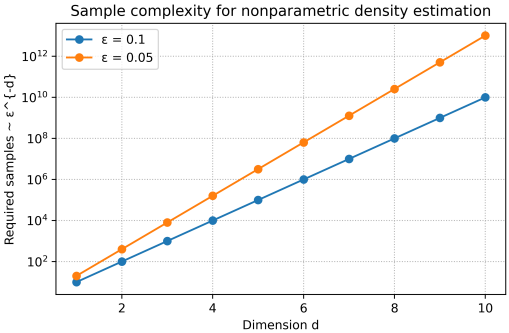
$$m = O\left(\epsilon^{-d}\right).$$

3. **Distance concentration:** For random vectors X_1, X_2:

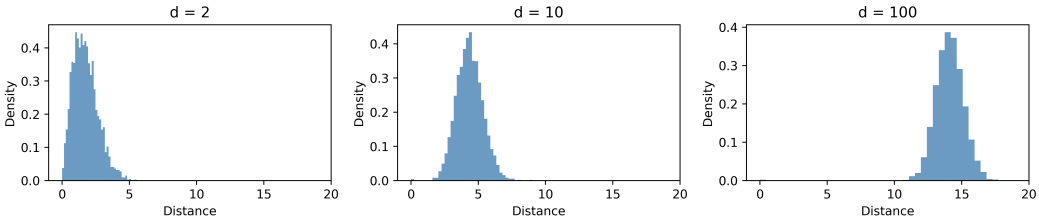
$$\frac{\|X_1 - X_2\|_{\text{max}} - \|X_1 - X_2\|_{\text{min}}}{\|X_1 - X_2\|_{\text{min}}} \xrightarrow{d \to \infty} 0.$$

The curse of dimensionality

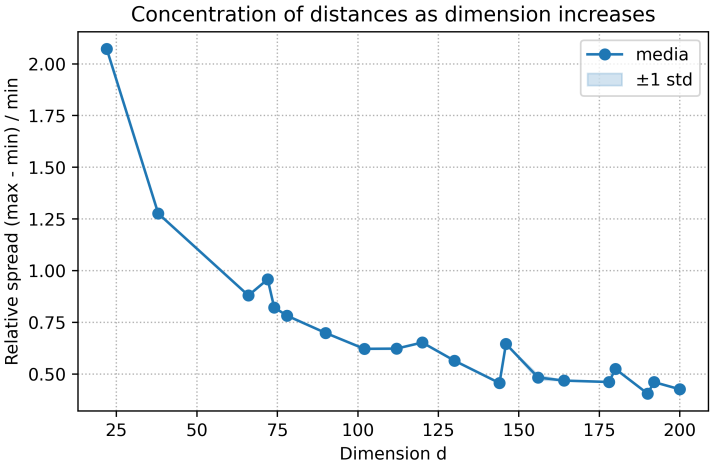




The curse of dimensionality



The curse of dimensionality



Practical strategies for mitigating the curse

1. **Dimensionality reduction:**
    - PCA (Principal Component Analysis).
    - Autoencoders.
    - t-SNE, UMAP.
2. **Feature selection:**
    - Filter methods (correlation, mutual information).
    - Wrapper methods (forward/backward search).
    - L1 regularization (Lasso).
3. **Underlying structure:**
    - Manifold hypothesis: high-dimensional data lives on low-dimensional manifolds.
    - Exploiting structure (sparsity, smoothness).

Learning with limited data

When m is small relative to d or problem complexity:

1. **Data augmentation.** Generate synthetic examples:
   - Transformations (rotation, noise, etc.).
   - GANs, generative models.
2. **Transfer learning.** Use knowledge from related tasks.
   - Pre-training on large datasets
   - Fine-tuning on specific data
3. **Few-shot learning.** Learn from very few examples:
   - Meta-learning.
   - Siamese networks, matching networks.
4. **Semi-supervised learning.** Exploit unlabeled data.

Deep neural networks challenge classical theory

- Millions of parameters (enormous d_{VC})
- Classical bounds predict terrible generalization
- In practice: **excellent generalization**

Modern explanations:

- **Implicit regularization:** SGD favors “flat’’ minima.
- **Double descent:** Non-monotonic generalization curve.
- **Neural Tangent Kernel:** Kernel perspective in overparameterized regime.
- **Lottery ticket hypothesis:** Effective small subnetworks.

Conclusions: Fundamental considerations in learning

$$\text{Capacity of } \mathcal{H} \uparrow \implies \begin{cases} R_{\text{emp}} \downarrow & \text{(better fit)} \\ \text{Generalization gap} \uparrow & \text{(risk of overfitting)} \end{cases}$$

Conclusions: Fundamental considerations in learning

- **No free lunch:** No universally best algorithm.
- **Inductive bias:** Necessary to incorporate domain knowledge.
- **Empirical vs theoretical:** Gap between theoretical bounds and practice.
- **Balance:** Model complexity, data quantity, regularization.

Theory provides: Intuition, design principles, warnings about pitfalls.